02/14/2022

Monday, February 14, 2022 11:29 AM

HELD-107

Starting Wednesday, 2/16.

Time : 11:30 – 12:20

Principles of Linear Programming

Annika has \$3000 available to invest, with which she buys shares of stock in the Bass Company and Forte Inc. (…) Determine the number of shares Annika should buy of each kind of stock in order to maximize the amount from dividends that she will receive at the end of the first year.

Let x = the number of shares of Bass stock

Let y = the number of shares of Forte stock

Let D = the total amount of dividends earned

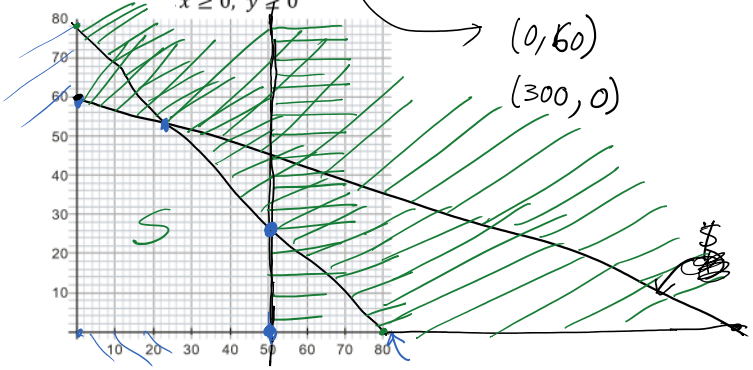
OBJECTIVE: Maximize $D = 3x + 4y$

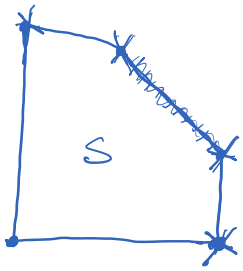
SUBJECT TO: $x + y \le 80$ Total number of shares

$10x + 50y \le 3000$ Total \$ invested

$x \le 50$ Limit on Bass stock

$x \ge 0, y \ge 0$





Can she buy 60 shares of Bass and 10 shares of Forte?

No, $x \leq 50$

Can she buy 10 shares of Bass and 65 shares of Forte? NO

$10 \times 10 + 65 \times 50 = 100 + 3250 = 3350 > 3000$

Can she buy 40 shares of Bass and 50 shares of Forte?

NO, $x + y = 90 > 80$

Can she buy 30 shares of Bass and 40 shares of Forte? YES.

$30 \times 10 + 40 \times 50 = 300 + 2000 = 2300 \leq 3000$

If Annika buys 48 shares of Bass and 0 shares of Forte, how much does she earn in dividends?

$3 \times 48 = 144$

If Annika buys 0 shares of Bass and 36 shares of Forte, how much does she earn in dividends?

$4 \times 36 = 144$

Solving Linear Programming Problems

Every linear programming problem has a feasible region associated with the constraints of the problem. These feasible regions may be bounded, unbounded or the empty set.

To find the solution (that is, where the maximum or minimum value occurs), we will use the two theorems below.

Theorem 1 If a linear programming problem has a solution, then it must occur at a vertex, or corner point, of the feasible set S, associated with the problem. Furthermore, if the objective function P is optimized at two adjacent vertices of S, then it is optimized at every point on the line segment joining these vertices, in which case there are infinitely many solutions to the problem.

Theorem 2 Suppose we are given a linear programming problem with a feasible set S and an objective function $P = ax + by$.

- **Case 1** If S is bounded, then P has both a maximum and a minimum value on S.
- **Case 2** If S is unbounded and both a and b are nonnegative, then P has a minimum value on S provided that the constraints defining S include the inequalities $x \geq 0$ and $y \geq 0$.
- **Case 3** If S is the empty set, then the linear programming problem has no solution; that is, P has neither a maximum nor a minimum value.

Example

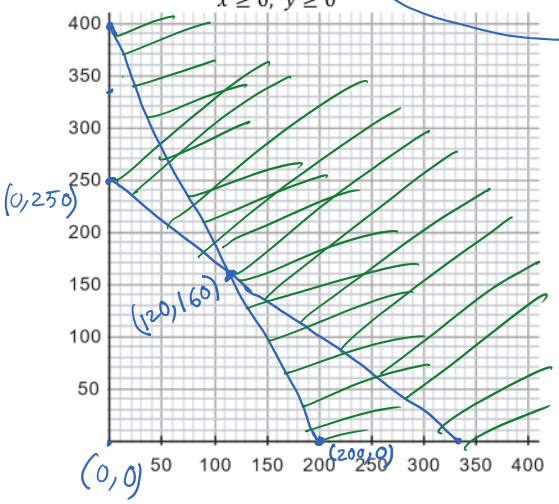
A company produces handmade skillets in two sizes, big and giant. (…) how many each size skillet should be made to maximize profits?

x = the number of big skillets produced
y = the number of giant skillets produced
P = the profits (in \$) from selling skillets
OBJECTIVE: Maximize $\boxed{P = 20x + 25y}$
SUBJECT TO:

$3x + 4y \le 1000$ Pounds of iron
$6x + 3y \le 1200$ Minutes of labor
$x \ge 0,\ y \ge 0$

$(0, 250)$
$(333\frac{1}{3}, 0)$

$(0, 400)$
$(200, 0)$



$3x + 4y = 1000$
$6x + 3y = 1200$ → RREF

$$\left[\begin{array}{cc|c} 3 & 4 & 1000 \\ 6 & 3 & 1200 \end{array}\right] \xrightarrow{RREF} \left[\begin{array}{cc|c} 1 & 0 & 120 \\ 0 & 1 & 160 \end{array}\right]$$

$x = 120$
$y = 160$

$P(200, 0) = 4000$

$P(0, 250) = 6250$

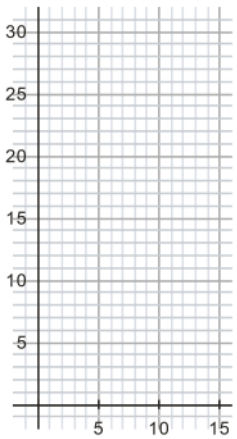
$P(120, 160) = 20 \times 120 + 25 \times 160 = 2400 + 4000 = 6400$

Example

A dietitian is to prepare two foods in order to meet certain requirements. (…) How many ounces of each type of food should be used in order to minimize the cost?

x = the number of ounces of food I
y = the number of ounces of food I
C = the cost (in \$) for the food
OBJECTIVE: Minimize $C = 0.25x + 0.15y$
SUBJECT TO:

$100x + 10y \geq 260$ Units of Vit. C
$40x + 80y \geq 320$ Units of Vit. D
$20x + 15y \geq 120$ Units of Vit. E
$x \geq 0,\ y \geq 0$



Example

A craftsman has 150 units of wood, 90 units of glue and 150 units of paint. (…) how many of each should be made to maximize the revenue?

x = the number of small picture frames produced

y = the number of large picture frames produced

C = the from selling picture frames (in dollars)

OBJECTIVE: Maximize $R = 175x + 400y$

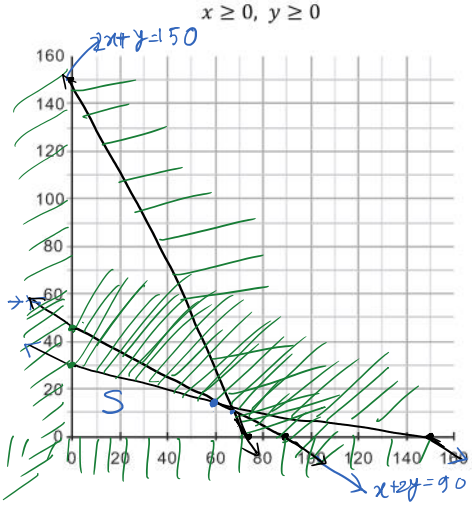
SUBJECT TO:

$1x + 5y \le 150$ Units of wood

$1x + 2y \le 90$ Units of glue

$2x + 1y \le 150$ Units of paint

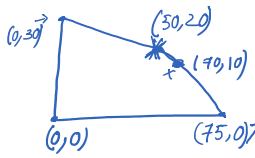
$x \ge 0,\ y \ge 0$



$$\begin{cases} x + 5y = 150 \\ x + 2y = 90 \end{cases} \rightsquigarrow x = 50,\ y = 20$$

$x + 2y = 90$

$2x + y = 150$ $\rightsquigarrow (70, 10)$



Vertex	$R = 175x + 400y$
(0,0)	0
(0,30)	12000
(50,20)	16750
(70,10)	16250
(75,0)	13,125

50 → small frames

20 → large frames.

Example

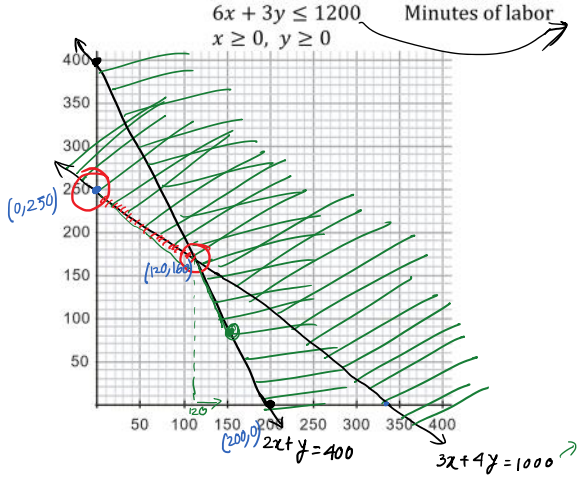
A company produces handmade skillets in two sizes, big and giant. (…) how many each size skillet should be made to maximize profits if big skillets have a profit of \$30 each and giant skillets have a profit of \$40 each?

x = the number of big skillets produced
y = the number of giant skillets produced
P = the profits (in \$) from selling skillets
OBJECTIVE: Maximize $P = 30x + 40y$
SUBJECT TO:

$3x + 4y \le 1000$ Pounds of iron
$6x + 3y \le 1200$ Minutes of labor
$x \ge 0,\ y \ge 0$

$2x + y \le 400$

$3x + 4y = 1000$
$2x + y = 400 \rightsquigarrow (120, 160)$



Vertex	$P = 30x + 40y$
(0,0)	0
(0,250)	10000
(200,0)	6000
(120,160)	10000

0 → Big Skillets
250 → Giant Skillets
120 → Big
160 → Giant.

$y = \dfrac{1000 - 3x}{4}$

$x \le 120$
$y = \dfrac{1000 - 3x}{4}$

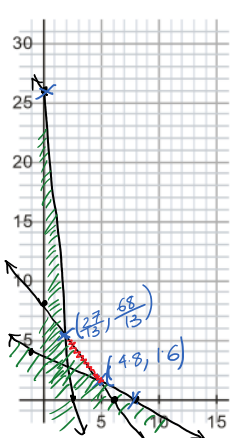
To maximize profit P,
we can make at most 120 big skillets
and we can make giant skillets
with the remaining resources.

Example

A dietitian is to prepare two foods in order to meet certain requirements. (…) How many ounces of each type of food should be used in order to minimize the cost if an ounce of food I costs 20 cents and an ounce of food II costs 15 cents?

x = the number of ounces of food I
y = the number of ounces of food I
C = the cost (in \$) for the food
OBJECTIVE: Minimize $C = 0.20x + 0.15y$
SUBJECT TO:

$100x + 10y \ge 260$ Units of Vit. C → $10x + y = 26$ → (0, 26), (2.6, 0)
$40x + 80y \ge 320$ Units of Vit. D → $x + 2y = 8$ → (8, 0), (0, 4)
$20x + 15y \ge 120$ Units of Vit. E → $4x + 3y = 24$ → (6, 0), (0, 8)
$x \ge 0,\ y \ge 0$

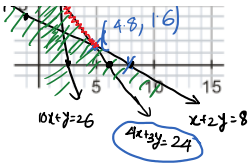


$C = 0.2x + 0.15y$, $7x + 3y$

Vertex	$C = 0.2x + 0.15y$
(0,26)	3.9
(8,0)	1.6
(4.8, 1.6)	1.2
(27/13, 68/13)	1.2

$-2x \to -2$
$\to -20$

The min cost \$1.2 occurs when



$\left(\frac{27}{13}, \frac{82}{13}\right)$ | 1.2

The min cost \$1.2 occurs when $\frac{27}{13} \le x \le 4.8$ oz of Food I and $\rightarrow y = -\frac{4}{3}x + 8$ oz of Food II.